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Proceedings of the Combustion Institute 33 (2011) 913–920
Proceedings of the Combustion Institute
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Experimental studies of the fundamental flame speeds of syngas (H_2/CO)/air mixtures

Nicolas Bouvet [a], Christian Chauveau [a,*], Iskender Gökalp [a], Fabien Halter [b]

[a] *Centre National de la Recherche Scientifique Institut de Combustion, Aérothermique, Réactivité et Environnement, 1C, Avenue de la Recherche Scientifique, 45071 Orléans Cedex 2, France*
[b] *Université d'Orléans, Institut PRISME, 8 rue Léonard de Vinci, 45072 Orléans Cedex 2, France*

Available online 9 September 2010

Abstract

Syngas laminar flame speed measurements have been carried out at atmospheric pressure and ambient temperature using spherically expanding flames. Mixture compositions ranging from 5/95% to 50/50% H_2/CO and equivalence ratios from 0.4 to 5.0 have been investigated. Both state-of-the-art linear and non-linear extrapolation methodologies have been tested and extracted laminar flame speeds have been compared to datasets available in the literature as well as computations using two leading kinetic mechanisms for syngas combustion. Syngas flame sensitivities to stretch have been characterized by extracting the corresponding Markstein lengths. In order to explain important disparities found for rich syngas flames among datasets of the literature, computations have been performed to quantify the possible extent of velocity reduction corresponding to small contents of iron pentacarbonyl.
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Keywords: Laminar burning velocity; Markstein length; Syngas; Spherical expanding flame; Stretch rate

1. Introduction

Synthetic gas mixtures, better known as "Syngas", have recently received a particular interest from the combustion community due to the increasing demand of the gas turbine industry to adapt dry low NO_x combustors for H_2/CO blend combustion. Among fundamental flame properties required to better understand "syngas burning", the fundamental flame speed is probably one of the most important since it provides detailed information on thermochemical processes governing the fuel combustion. It is therefore extensively used for the validation of combustion models.

Several fundamental flame speed studies of syngas blends have been reported in the literature including measurements performed on conical [1–3], counterflow [4] and outwardly expanding flames [5–9]. If the former configuration is known to suffer from stretch effect, the two latter allow for its systematic subtraction using adapted extrapolation procedures. Recent improvements for the outwardly propagating flame processing algorithm include both robust linear [10] but also non-linear extrapolation methodologies specifically developed for higher hydrocarbon fuels [11]. This non-linear approach provides increased accuracy for flames with strong non-linear answers at higher stretches

* Corresponding author. Fax: +33 238 25 78 75.
E-mail address: chauveau@cnrs-orleans.fr (C. Chauveau).

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doi:10.1016/j.proci.2010.05.088

and by avoiding biasing effects inherent to linear extrapolation procedures.

Although several syngas flame speed investigations have been carried out in the past, just a very few covered a large range of equivalence ratios, including both ultra-lean to very rich mixtures. Among these ones, important scatters are still remaining, as pointed out in a recent syngas combustion kinetics review by Chaos and Dryer [12]. The objectives of the present investigation are therefore to accurately determine fundamental flames speed of various syngas blends using the outwardly propagating flame configuration and both linear and non-linear processing methodologies. The results will be compared to datasets of the literature as well as computations using leading syngas kinetic mechanisms. Discrepancies between results obtained with both linear and non-linear procedures will be characterized. Disparities observed between available flame speed datasets will be numerically analyzed in the light of a potential influence of carbonyl compounds, known to be strong flame inhibitors [12,13].

2. Experimental approaches

2.1. General configuration and experimental procedure

The experimental apparatus and diagnostic arrangement used in the present investigation are similar to those presented in Ref. [10]; they will be briefly described here. The combustion vessel is a 24.32 l stainless steel cylindrical chamber (160 mm ID, 300 mm height). Two sharpened-edge tungsten electrodes linked to a conventional capacitive discharge ignition system are used to provide the ignition energy. The electrode plane is slightly tilted with respect to the plane of observation to avoid the recording of disturbances inherent to the ignition event. The spark gap can be varied and was usually set between 1 and 2 mm. The volumes of each mixture component (H_2/CO/air) are sequentially injected in the combustion chamber thanks to dedicated mass flow controllers digitally piloted via a RS232 serial port. Gas volumes delivered by the flow controllers were periodically checked with wet meters (measuring uncertainty of 0.5%) and were found to be within 1.2% of the input values, allowing an equivalence ratio precision of $\pm 1.4\%$. The pressure and the temperature inside the chamber are measured using a piezoelectric transducer and a type K thermocouple, respectively. A fan located inside the vessel ensures a homogeneous mixing of reactants prior to ignition. Specific efforts were made to render the entire gas supply network compatible with CO, as further detailed in Section 4.3.

The optical access to the chamber is provided through two opposite portholes. A continuous Stabilite 2017 argon ion laser (6 W, $\lambda = 457.9$–514.5 nm) along with two planoconvex lenses ($f_1 = 25$ mm; $f_2 = 1000$ mm) are used to create the parallel light beam that crosses the combustion chamber. A transparent screen located at its back allows for the display of the shadowgraph images. Shadowgrams of outwardly propagating flames are recorded using a high speed CMOS APX camera operating at 6000 and 15,000 frames/s for 512×512 and 256×256 pixel size frames, respectively. The spatial resolution achieved is close to 135 μm/pixel in all cases. The entire setup is shown in Fig. 1.

The initial pressure and temperature conditions were kept in the ranges of 1.04 ± 0.035 bar and 295.3 ± 3.7 K, respectively. Ignition is initiated few minutes after complete chamber filling in order to make sure that quiescent flow conditions were achieved. The ignition sequence was then triggered and the flame front propagation evolution was recorded using the synchronized high speed camera.

During the present investigation, experimental runs were repeated three times in average. The corresponding standard deviations for the flame propagation speed were found to be in a 0.1–7.8 cm s^{-1} range. They are plotted in the form of vertical error bars in the graphs presented herein.

2.2. Processing steps

The flame propagation observed on the recorded movies demonstrates a spherical evolution for all flame radii. A conventional assumption to simplify the flame front processing step is that the development of such flames is perfectly spherical and hence, flame fronts can be fitted by the equation of a circle. This approach has been adopted by Tatouh and coworkers in Ref. [10]. Their flame edge detection program is here applied to the present shadowgraph images. This program is successively performing: (i) a background subtraction to enhance the flame front detection;

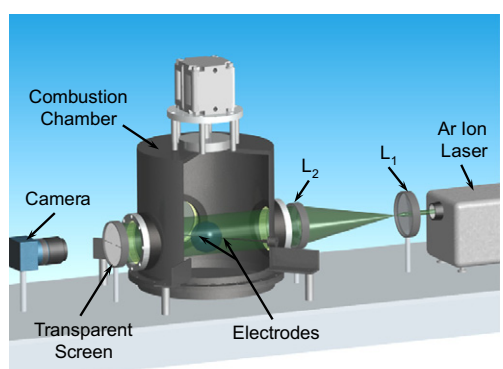

Fig. 1. Schematic of the combustion chamber and shadowgraph system.

(ii) a flame edge detection based on the maximum intensity of the flame front; (iii) the luminous edge best fit by performing an optimization algorithm based on the minimization of the distance between detected points and points belonging to the potential solution. Figure 2 shows a spherical flame front with its superimposed best fit (red circle). These processing steps are repeated for all recorded shadowgrams allowing for the extraction of the temporal evolution of the flame radius. Examples of such evolutions are given in Fig. 3 for various equivalence ratios. It is seen that for the lean and moderately rich cases (ER = 0.6, 1.0, 2.0) linear trends are observed. For the richest case (ER = 5.0), a slower flame radius increase is observed during the initial phase of the flame propagation (up to $t = 7$ ms) but a linear trend is recovered in the later stages of the flame evolution. This observation is coherent with experimental and numerical results of Chen et al. [14] and Kelley et al. [15] for which it was reported that, for mixtures with large Lewis numbers, the ignition energy as well as initial transient flame evolution have a significant impact on early flame trajectories.

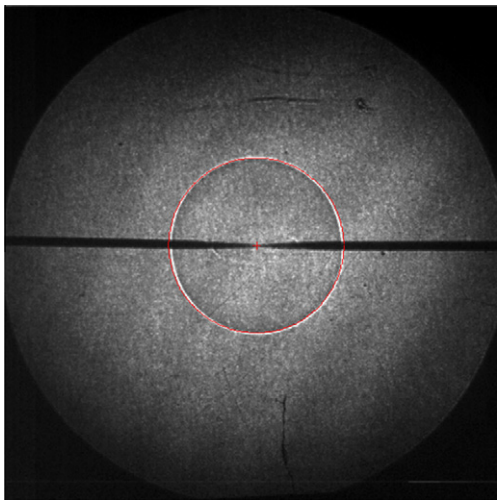

Fig. 2. Illustration of the flame front edge detection (in red, circle best fit). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

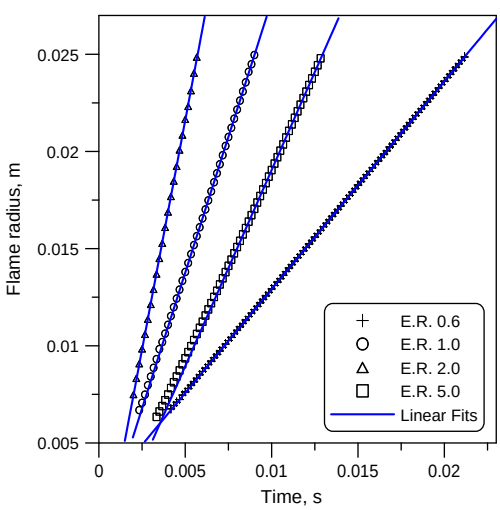

Fig. 3. Temporal evolution of flame radii for different equivalence ratios (10% H_2–90% CO). Linear fits performed on the last 4 ms of the radius evolutions.

The classical outwardly propagating flame approach assumes that a linear relationship exists between the stretched propagation flame speed S_b and the applied stretch rate K. Such a relation was originally suggested in the pioneer works of Markstein [16] and largely discussed in the later theoretical and numerical studies [17,18]:

$$S_b = S_b^0 - L_b K \tag{1}$$

with S_b^0, the unstretched propagation flame speed and L_b, the Markstein length for burned gases. The stretched propagation flame speed is related to the flame front radius as follows:

$$S_b = dr/dt \tag{2}$$

The flame stretch, solely governed by flame motion and curvature effects, is expressed for a spherically symmetric flame as follows:

$$K = \frac{1}{A}\frac{dA}{dt} = \frac{2}{r}\frac{dr}{dt} \tag{3}$$

with A, the flame surface. Temporal evolutions of the flame radii, as shown in Fig. 3, can be used to yield the (K; S_b) pairs and perform appropriate extrapolation procedures to yield both the unstretched flame propagation speed S_b^0 and the Markstein length of burned gases L_b. In the case of a flame expansion at constant pressure, the laminar burning velocity S_u^0 is then deduced from the unstretched propagation flame speed using the burned to unburned gas density ratio:

$$S_u^0 = (\rho_b/\rho_u) S_b^0 \tag{4}$$

with ρ_b and ρ_u the burned and unburned gas densities, respectively.

A crucial point of the data processing step is the methodology used to extract both S_b^0 and L_b. In the present investigation, two different approaches have been considered:

- The recent linear extrapolation procedure proposed by Tahtouh and coworkers [10] based on the exact analytical solution of the differential equation of the flame front temporal evolution (5):

$$\frac{dr}{dt} = S_b^0 - 2L_b \frac{1}{r}\frac{dr}{dt} \tag{5}$$

where (5) is obtained combining expressions (1)–(3). It has been demonstrated that this methodology yields an enhanced robustness and improved accuracy as compared to conventional linear processing schemes. Ref. [10] should be consulted for additional information on the exact analytical formulation of the solution of Eq. (5) as well as the extraction of the parameters of interest S_b^0 and L_b.

- The non-linear extrapolation methodology described by Kelley and Law in Ref. [11], based on the non-linear evolution equation for the stretch-affected flame propagation speed originally proposed by Ronney and Sivashinsky [19]:

$$\left(\frac{S_b}{S_b^0}\right)^2 \ln\left(\frac{S_b}{S_b^0}\right)^2 = -\frac{2L_bK}{S_b^0} \tag{6}$$

The development of this approach was motivated by the apparent non-linearity of flame propagation velocities at higher stretch. This methodology was shown to provide an enhanced accuracy for the determination of fundamental flame speeds as compared to the conventional linear approaches, especially for nonequidiffusive mixtures. Fitting algorithms developed in Ref. [20] are used in the present investigation.

3. Numerical approach

The Sandia flame code PREMIX [21] along with the CHEMKIN 2 subroutines [22] and the transport property subroutines [23] are used to compute 1D unstretched laminar flame speeds. Two leading kinetic mechanisms for syngas combustion are considered: these are the recently developed mechanisms of Li et al. [24] (84 reactions among 21 species) and Sun et al. [7] (33 reactions among 16 species). Both aforementioned C1 mechanisms, integrating updated rate constants and thermochemical parameters, have been extensively validated, not only against H_2/CO laminar flame speed datasets, but also ignition delays and temperature as well as flow reactor species profiles. To further investigate the influence of iron pentacarbonyl on syngas flames, the kinetic subset developed by Rumminger et al. [25] (8 iron-containing species, 55 reactions) was added to the Li et al. [24] kinetic scheme. Calculations were performed with the original thermodynamics and transport data provided in Refs. [7,24,25] and multi-component diffusion and thermal diffusion (Soret effect) options were included. All computations were done using the premix continuation option allowing, in a first pass, for the expansion of the computational domain, followed by a progressive decrease of the GRAD and CURV options, ultimately set to 0.1. As such, final solutions were generally obtained on more than 120 mesh points. Note that for most cases, calculated velocities for final grids (≈140–180 mesh points) are well within a 2 cm s^{-1} interval as compared to intermediate solutions (≈70–100 mesh points). This level of convergence is sound if accuracies of conventional laminar flame speed techniques are considered.

4. Results and discussion

Unless otherwise stated, all measurements presented in this section were obtained using the non-linear methodology.

4.1. Syngas laminar flames speeds and Markstein lengths

The summary of flame speed measurements for all investigated syngas/air mixtures is presented in Fig. 4 along with the predictions calculated with the Li et al. mechanism [24]. It is observed that calculations generally overpredict the measurements up the curve extrema. The agreement is good for both 5/95% and 10/90% H_2/CO mixtures for the very rich branch but measurements are higher than predictions for both 25/75% and 50/50% H_2/CO cases. The maximum difference is obtained for the 25/75% composition at ER = 5 with 17.6 cm s^{-1}. Measurements for each individual composition are further compared to available results of the literature in the outwardly propagating flame configuration in Figs. 5–8.

For the 5/95% H_2/CO case (Fig. 5), the agreement is good for all series up to ER = 2.5. Note that on the ER range 0.4–2.5, predictions given by the Li et al. [24] and Sun et al. [7] mechanism seem to slightly overpredict all measurements. However disparities can be noticed on the rich

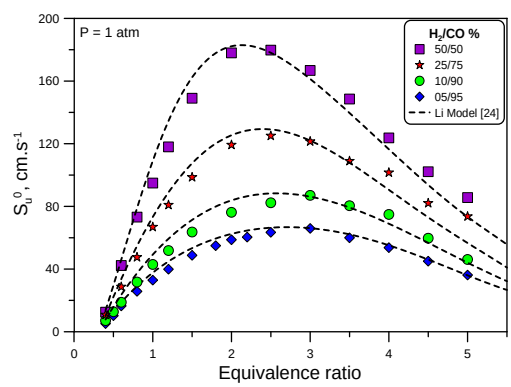

Fig. 4. Results summary for laminar flame speeds of syngas mixtures.

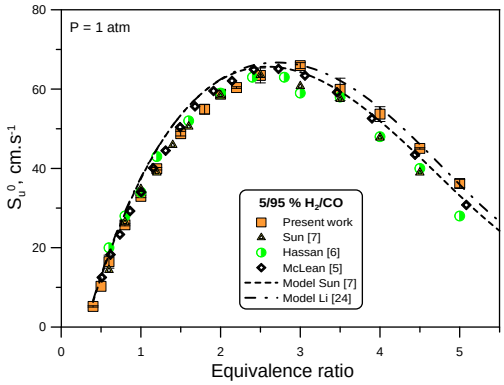

Fig. 5. Laminar flame speeds of 5/95% H_2/CO mixtures. Comparison with experimental datasets and numerical predictions.

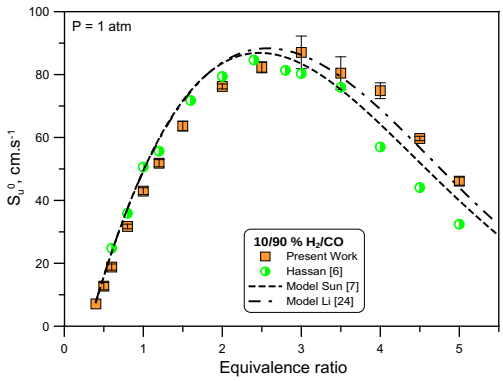

Fig. 6. Laminar flame speeds of 10/90% H_2/CO mixtures. Comparison with the experimental dataset and numerical predictions.

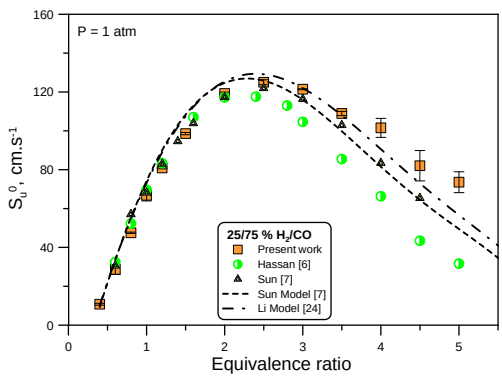

Fig. 7. Laminar flame speeds of 25/75% H_2/CO mixtures. Comparison with experimental datasets and numerical predictions.

side (ER > 2.5), especially between velocities provided in Refs [6,7] and those given by the present work. Measurements reported by McLean and coworkers [5] lay in an intermediate range. Notice here that results of the present investigation show an excellent agreement with computations using the mechanism of Li et al. [24] for the very rich cases (ER = 4.5–5.0). A maximum discrepancy of 8.2 cm s^{-1} is obtained at ER = 5.0 between the present results and those of Hassan et al. [6].

Disparities are further increased among experimental series for the 10/90% H_2/CO case (Fig. 6). If present results are still overestimated on the lean side up to ER = 2.5, they do find a good agreement with predictions of the Li et al. mechanism [24]. Measurements of Hassan and coworkers [6] lay significantly lower than the present results with a maximum difference of 20.1 cm s^{-1} obtained at ER = 4.0.

The same observations hold for the 25/75% H_2/CO case (see Fig. 7) with an excellent agreement among experiments up to ER = 2.0. Above, important differences can be seen for the three sets of measurements including the present one. Lowest velocities are found by Hassan and coworkers [6] and highest are given by the present investigation. Notice that results of the present work are, for the rich branch, higher than predictions obtained with the Li et al. [24] mechanism. At ER = 4.5, we have $(S^0_{u\mathrm{HASSAN}} = 41\ \mathrm{cm\ s^{-1}}) < (S^0_{u\mathrm{SUN}} = 65.4\ \mathrm{cm\ s^{-1}}) < (S^0_{u\mathrm{PRESENT\ WORK}} = 82.1\ \mathrm{cm\ s^{-1}})$.

Figure 8 shows the results obtained for the 50/50% H_2/CO mixtures. A good agreement can be seen for the very lean side, but some deviations occur for ER = 1.0–1.5. On this range, present measurements give the lowest velocities as compared to Prathap et al. [8] and Sun et al. [7] datasets. The rich branch is still characterized by important disparities, especially between the present results and the data points of Hassan and coworkers [6] for which a 46.1 cm s^{-1} difference at ER = 4.5 can be observed. However, an excellent agreement is found between results of the

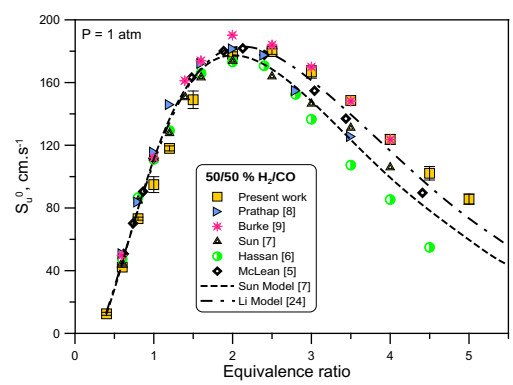

Fig. 8. Laminar flame speeds of 50/50% H_2/CO mixtures. Comparison with experimental datasets and numerical predictions.

present work and the recent measurements of Burke et al. [9]. Both are however largely underpredicted by the kinetic mechanism of Sun and coworkers [7] and, to a lesser extent, by the mechanism of Li et al. [24].

It is important to underline that observed disparities for flame velocities of rich syngas mixtures are out of the range of the experimental accuracy and repeatability: (i) the direct methodology accuracy i.e., the expected accuracy for a single measurement, is closely related to the extrapolation procedure employed to yield S_b^0. It is expected to be on the order of few cm s^{-1}; (ii) the repeatability, deduced from data series performed for the same experimental conditions, is found to be, in average, within a ± 2 cm s^{-1} range for the present experiments. As mentioned earlier, the largest deviation is ± 7.8 cm s^{-1} for the 25/75% H_2/CO mixture at ER = 4.5.

Markstein lengths for the different investigated mixtures are presented in Fig. 9. Interestingly, all flames generally display a fairly poor sensitivity to stretch. The only exception is met for the 50/50% H_2/CO mixture where L_b assumes a sign inversion with a very sharp decrease in the ultra-lean side ($L_b = -2.9$ mm at ER = 0.4). This trend suggests that strong nonequidiffusion effects (mixture Lewis number $Le \ll 1$) affect the propagation of these ultra-lean flames. This behavior is further confirmed by experimental works available in the literature (see Fig. 10). If an overall good agreement is found with results provided by Brown et al. [26], Hassan et al. [6] and Prathap et al. [8] on the 0.4–2.5 equivalence ratio range, disparities can be noticed among Markstein lengths of very rich mixtures (ER > 3.0). These discrepancies might be explained by differences in processing procedures adopted by the different authors, especially the choice of the useful experimental data range used for the flame speed extrapolation. Recent theoretical analysis [14] indeed revealed that this range could be significantly narrowed for mixtures with large Lewis numbers, for which the initial ignition energy has a considerable impact on early flame trajectories. On the other hand, it is important to recall that, for the outwardly propagating flame configuration, the Markstein length is quasi-independent of the chosen isotherm at which it is evaluated [27], which eliminates a potential source of bias among different experimental investigations.

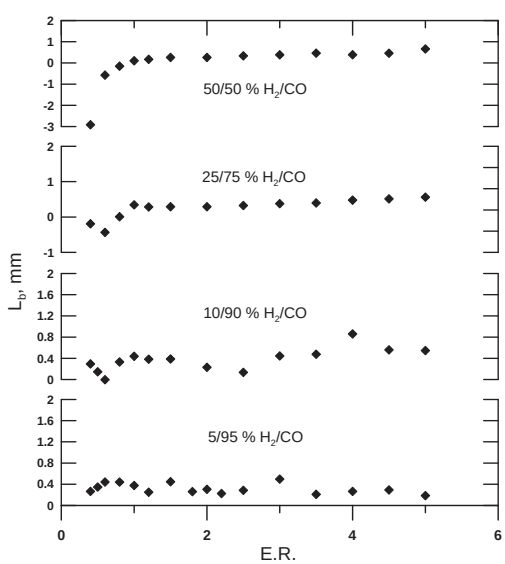

Fig. 9. Comparison of Markstein lengths for investigated syngas mixtures.

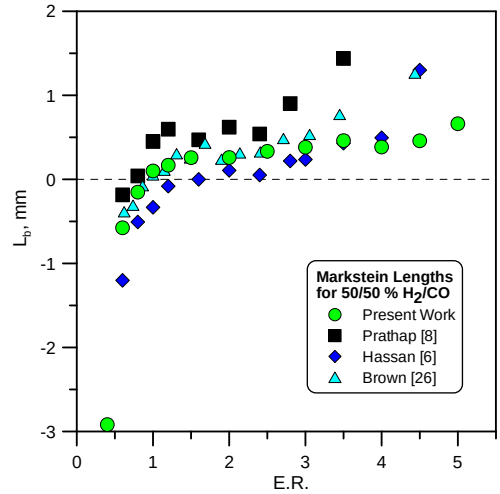

Fig. 10. Comparison of Markstein lengths for the 50/50% H_2/CO mixtures and experimental datasets from the literature.

4.2. Linear versus non-linear extrapolation methodologies

In order to characterize potential deviations between the linear and non-linear extrapolation methodologies used in the present investigation, results using both approaches have been compared for the 25/75% and 50/50% H_2/CO syngas mixtures. Three methodologies are considered for the comparisons: (i) the non-linear formulation (*NL*) proposed by Kelley and Law [11] (see Eq. (6)), (ii) the linear formulation (*L*) of Tahtouh and coworkers [10] with an initial flame radius Rf_{initial}, defined as the first flame front radius considered in the processing procedure, fixed at 8 mm; (iii) the linear method (*LC*) of Tahtouh and coworkers [10] following the criterion developed in ref. [20] for the choice of an adapted Rf_{initial}:

$$|L_b/Rf_{\text{initial}}| < 0.18 \quad (7)$$

This empirical criterion recommends that a larger initial flame radius should be used for mixtures with larger Markstein lengths in order to ensure that the linear approach is still valid as compared to the non-linear one. It was shown [28] that almost no difference can be observed between the three approaches on the ER range 0.4–4.0. Some discrepancies arise for laminar flame speeds of very rich syngas flames (ER = 4.5 and 5.0) with a maximum offset between the non-linear and linear (no criterion) approaches on the order of 8 cm s^{-1} (see Table 1). Notice here that laminar flame speeds $S^0_{u,LC}$, extracted according to the criterion (7), slightly overestimate non-linearly extracted values, suggesting that the reference ratio (7) should be decreased to recover identical flame speeds with linear methodologies. Since experimental studies of Refs. [6,8,26] mention Rf_{initial} on the order of 5 mm, it is interesting to notice that (7) leads to $|L_b| < 0.9$ mm and thus Markstein lengths and flame speed measurements provided by Ref. [8] at ER = 3.5 and Refs. [6,26] at ER = 4.5 should be carefully considered.

4.3. Potential carbonyl compound influence

A potential source of discrepancy among available flame speed datasets for the rich syngas mixtures presented in Figs. 6–8 could be explained by the presence of unwanted carbonyl compounds in the H_2/CO blends of interest. As recalled earlier, these compounds are known to have strong inhibition effects on flames. For the present study, important modifications of the experimental apparatus were engaged in order to minimize the possible online formation of $Ni(CO)_4$ and $Fe(CO)_5$: (i) the entire gas supply system is made of Teflon tubing, (ii) the conventional manometer used for combustible gases has been replaced by a SWAGELOK KCY two-stage regulator, especially customized for CO compatibility (body material: brass CW721R, seat/seal material: PCTFE). Since such information is currently not available for experimental setups described in the literature, it is therefore of prime interest to get an estimation of the corresponding flame speed reduction for realistic carbonyl contamination concentrations.

For the present computations, the iron pentacarbonyl kinetic subset of Rumminger et al. [25] has been added to the mechanism of Li and coworkers [24] in the manner of Ref. [12]. Different hypothetical levels of $Fe(CO)_5$ ranging from 50 to 200 ppm in CO have been tested. Results for the 10/90% and 50/50% H_2/CO mixtures are presented Figs. 11 and 12, respectively. It can be seen that for both cases, flame inhibition effects are essentially effective on the rich side of the flame speed curves. If a concentration of 100 ppm of $Fe(CO)_5$ could indeed explain the large disparities observed between the present measurements and those of Hassan and coworkers [6] for the 10/90% H_2/CO mixture (Fig. 11), the extent of flame speed reduction with $Fe(CO)_5$ contents up to 200 ppm is limited for the 50/50% H_2/CO flame case (Fig. 12). For instance, computations indicate a flame speed decay on the order of 6 cm s^{-1} for ER = 3.5 and 200 ppm of $Fe(CO)_5$, while experimental disparities span a 40 cm s^{-1} interval. This clearly suggests that, for syngas mixtures with high H_2 contents, the important flame speed scatter observed at

Table 1
Extrapolation methodology comparison (in cm s^{-1}).

H_2/CO	ER	$S^0_{u,L}$	$S^0_{u,LC}$	$S^0_{u,NL}$	$S^0_{u,L} - S^0_{u,NL}$
25/75%	4.5	89.8	83.7	82.1	7.7
	5.0	78.2	75.8	73.6	4.6
50/50%	4.5	110.2	105	102.1	8.1
	5.0	93	90.1	85.6	7.4

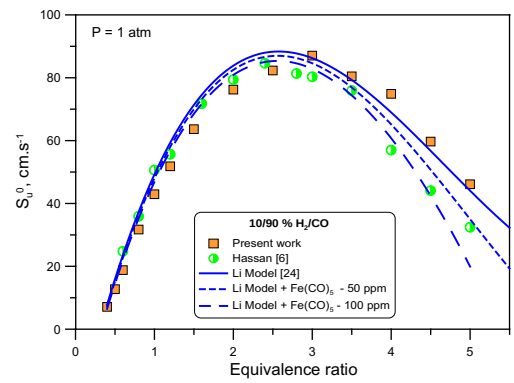

Fig. 11. Laminar flame speeds for the 10/90 % H_2/CO blend compared with model predictions incorporating two levels of iron pentacarbonyl contamination: 50 and 100 ppm.

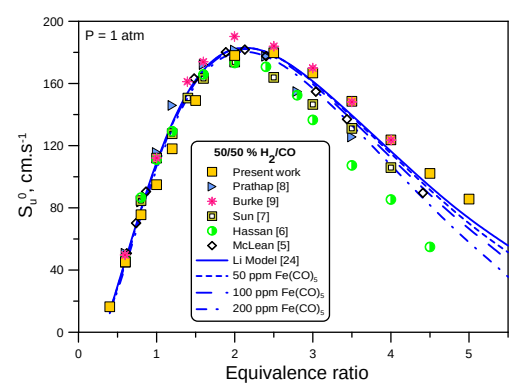

Fig. 12. Laminar flame speeds for the 50/50 % H_2/CO blend compared with model predictions incorporating three levels of iron pentacarbonyl contamination: 50, 100 and 200 ppm.

higher equivalence ratios cannot be solely explained by potential impurities related to CO sources.

5. Conclusion

Laminar flame speeds of H_2/CO/air mixtures were accurately measured for a wide equivalence ratio range at atmospheric pressure. Large discrepancies exit among available literature datasets for rich syngas flames. The present measurements give the highest trends and are best predicted by the mechanism of Li and coworkers [24]. Excellent agreement was found between recent experimental data of the literature and the present results for the fastest flames. In order to understand the reasons of the disparities of the earlier experimental results, the data reduction methods were tested and compared. The use of the non-linear methodology is recommended for the specific case of very rich mixtures, with higher H_2 contents. Furthermore, it was found numerically that flame inhibition effects of iron pentacarbonyl could explain, to some extent, the discrepancies among experimental results. Further studies are now required to experimentally confirm and quantify the corresponding velocity reductions, for instance by investigating rich syngas flames doped with small amounts of carbonyl compounds.

Acknowledgments

Dr. Bouvet was supported by a joint grant from the CNRS and the Conseil Regional Centre. Experiments were carried out at PRISME-Université d'Orléans. Help of the EPM team is gratefully acknowledged. The authors thank Dr. H. Sun and Dr. G. Linteris for providing the input files of their respective kinetic mechanisms.

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